

MODEL QUESTION

Signature.....

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/ TECHNOLOGY.

INDUSTRIAL AUTOMATION AND MECHATRONICS

[Maximum marks: 100]

[Time 3 hrs]

PART- A

marks

- I. Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define the term 'Automation'
 2. Define the term 'stability' of a transducer.
 3. Define 'Dead band' of a transducer.
 4. Draw the symbol of a 2 port 2 position poppet valve
 5. Draw the symbol of ladder, where input as contacts not closed until input
- (5 x 2=10)

PART – B

Answer any five of the following. Each question carries 6 marks

- II.
1. Explain the elements of product design.
 2. Distinguish displacement, position and proximity sensor
 3. Explain the working of eddy current proximity sensor
 4. Explain the characteristic of diode
 5. Explain the working principle of D C Motor .
 6. Explain the ladder logic diagram
 7. Draw the block diagram of 8051 microcontroller.
- (5 x 6= 30)

PART C

(Answer one full question from each module, Each question carries 15 marks))

Module I

- III.
- | | |
|--|---|
| 1. Explain the different types automation | 4 |
| 2. Discuss the measurement system with the help of a block diagram | 4 |
| 3. Distinguish between Traditional and Mechatronic design | 7 |

OR

- IV
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|--|---|
| 1. Explain closed loop control system with the help of a sketch. | 8 |
| 2. Discuss the advantages and disadvantages of automation | 7 |

Module II

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|-----|---|---|
| V. | 1. Explain non linearity error with the help of sketches | 8 |
| | 2. Discuss force measurement with the help of strain gauge load cell. | 7 |
| | OR | |
| VI. | 1.Explain bouncing and debouncing of mechanical switches. | 4 |
| | 2.Discuss any one software method for debouncing of mechanical switches | 4 |
| | 3.With the help of sketches explain the working of incremental encoder | 7 |

Module III

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| VII. | 1. Explain the different types of D C Motor | 7 |
| | 2. Explain solid state switches: a. diode b. triac | 8 |

OR

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| VIII. | 1. With the help of sketches explain the various types of process control valve | 8 |
| | 2. Differentiate Hydraulic and pneumatic systems | 7 |

Module IV

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| IX. | 1. Explain the selection of PLC. | 7 |
| | 2. Explain timed switch in mechatronic design | 8 |

OR

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| X. | 1. Discuss the Fault finding techniques in microprocessor based system | 7 |
| | 2. Explain bathroom scale in mechatronic design | 8 |
